NE 235 – Nuclear Reactor Operations

Fall Semester – 2 Credits – S/U

Instructor: Dr. Colby Fleming – Dept. of Nuclear Engineering
Office: BU 2120; Email: ncsorrel@ncsu.edu
Lecture: Tuesday 8:30-10:20AM in Lampe Drive, Room 214
Lab: Thursday 8:30-10:20AM in Burlington Labs, Room 2111
Office Hours: *Individual appointments as needed*
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Course Description:

This course presents the basic nuclear engineering theory pertaining to fission reactor operations, and discusses the multiple systems integral to the functioning of the PULSTAR nuclear plant. The impacts of cyber security considerations on reactor safety, security, and control system instrumentation are presented. Lecture content covers both the theoretical and practical elements of reactor operations, with laboratory sessions providing hand on opportunities for students to apply their knowledge and operate the reactor plant.

The knowledge and hands-on experience student's gain will serve as a solid base that will facilitate understanding of advanced nuclear reactor theory and engineering course work. Qualified students may take additional training to stand for the Reactor Operator Licensing Exam as administered by the U.S. Nuclear Regulatory Commission.

Course Objectives:

1. Students will learn basic theory behind fission nuclear reactors including key modes of interaction, fission reactor operation, reactor safety in the modern era, and basic health physics.
2. Students will gain a strong physical understanding of nuclear plant operations. From the second week of the semester, they will be involved in hands-on training in learning to operate the PULSTAR nuclear research reactor and all of its associated systems.

   Students will:
   a. Demonstrate an understanding of the basic underlying principles of nuclear reactor operations.
   b. Perform PULSTAR facility pre-startup checks, reactor startups and shutdowns, and steady state reactor operations.
   c. Apply theory in explaining observed operating characteristics of the PULSTAR reactor.

Lecture Topics *(see Course Schedule on Page 5)*:
1. Chart of the Nuclides, Nuclear Reactions, Radioactive Decay Modes
2. Reactor Components and Materials, Reaction Rates, Fission Process
3. Effective Multiplication Factor and the Neutron Life Cycle, Reactivity and Criticality
4. Health Physics & Radiation Safety
5. Cyber Informed Engineering for Nuclear Systems.

6. Sub-Critical Multiplication & 1/M Approach to Criticality
7. Delayed and Prompt Neutrons and the Delayed Neutron Fraction
8. Inhour Equation, Reactor Period, and Exponential Power Equations
9. Reactivity Coefficients including Moderator Temperature, Voids, Doppler Feedback, and Fission Product Poisons.
10. Reactivity Balances, Excess Reactivity and Shutdown Margin
11. Neutron Flux Distribution and Power Density, Heat Balance & Power Calibration
12. Special Topic – Chernobyl Accident

Laboratory Sessions *(see Course Schedule on Page 5)*:

1. Introduction to PULSTAR Systems
2. Health Physics Instrumentation & Radiation Survey
3. Reactor Startup
4. 1/M Approach to Criticality
5. Control Rod Calibration
6. Power Defect Measurement
7. Axial Power Profile Measurement
8. Heat Balance and Power Calibration

Restrictive Statement:

This course does not count towards Nuclear Engineering Bachelor's degree requirements.

Textbooks:

The NE235 Reactor Operator Training Manual and all course content is available online through Moodle. No purchases of textbooks are required.

Grading:

This course is Pass-Fail (S/U) only. An overall grade point average of 70% (C-) or better is required to obtain a "Satisfactory" the course. Grades will be given for all homework assignments (25%), quizzes (25% ea. x 2 = 50%), and the final written exam (25%).

Policies on absences and scheduling makeup work.

Attendance is mandatory for all lectures and laboratory sessions. Absences and late or missed assignments can be excused for circumstances defined in NCSU's Attendance Policy (https://policies.ncsu.edu/regulation/reg-02-20-03-attendance-regulations/).
Any arrangements for making up missed class or laboratory work must be made with the instructor - in advance for non-emergency absences and immediately upon your return to class for emergency absences. In all cases, try to contact the instructor prior to an absence if possible.

Policy on late assignments:

Homework assignments will be due at the beginning of the class period on the due date. Late assignments will not be accepted without an approved excuse as defined in NCSU's Attendance Policy (see web-link above).

Academic Integrity:

Students are expected to adhere to the guidelines for academic integrity as outlined in the NCSU Student Code of Conduct (https://policies.ncsu.edu/policy/pol-11-35-01/). It is the instructor's understanding and expectation that the student's signature on any test or assignment means that the student neither gave nor received unauthorized aid. Cheating and plagiarism will result in the loss of credit for the entire test or assignment in question.

Students with Disabilities:

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with Disability Services for Students at 1900 Student Health Center, Campus Box 7509, 515-7653. For more information on NC State's policy on working with students with disabilities, please see http://dso.dasa.ncsu.edu/.

Laboratory Safety:

Laboratory sessions will be conducted in the PULSTAR Nuclear Reactor Facility. All federal and state safety regulations and in-house procedures will be followed. Students will undergo a radiological safety orientation prior to carrying out laboratory activities in the reactor bay.

End of Semester Class Evaluations:

Online class evaluations will be available for students to complete during the last two weeks of class. Students will receive an email message directing them to a website where they can login using their Unity ID and complete evaluations. All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any particular instructors.

Qualifications for Licensing:

The selection process for licensing spots is competitive. The number of open RO spots per year will vary due to budget constraints. The number of open spots is typically between one and three, but no minimum number is guaranteed.

To be considered for training in the spring semester to take the NRC Reactor Operator licensing exam, an RO candidate must meet the following minimum criteria:

- Must be available weekly for training activities (schedule TBA) in both the fall and spring semesters.

- Must receive an A- grade (90%) or better in NE235.
- Must have a $\geq$ 3.0 GPA in major and overall.
- Must be a Freshman or Sophomore. Upperclassmen and Graduate students will not be licensed.
- Must be able to serve as a licensed operator for a period of at least one year prior to graduation.
- Must be able to pass a medical physical examination per NRC requirements.
- Must be dependable and mature.

NE235 - NUCLEAR REACTOR OPERATIONS – FALL 2022 COURSE SCHEDULE*

Week #	Lecture *(Tuesdays 8:30-10:20)*	Materials	Assignment	Laboratory Session
1	**08/23 – Lect. #1**: Introduction to the Chart of the Nuclides – Nuclear Reactions, Radioactive decay modes	Reactor Theory Chapter 1: §1.1 – §1.3 + PDF Files	HW#1&2 (due 9/6)	**08/25:** PULSTAR Systems Intro
2	**08/30 – Lect. #2:** Reactor Components and Materials, Nuclear Reaction Rates and Fission Process	Reactor Theory Chapter 1: §1.1 – §1.3 + PDF Files	HW#1&2 (due 9/6)	**09/01:** Lab #1 - PULSTAR Systems Walkdown
3	**09/06 – Lect. #3:** Health Physics & Radiation Safety	PDF Files	HW#3 (due 9/13)	**09/08:** Lab #2 - HP Instruments
4	**09/13 – Lect. #4:** Effective Multiplication Factor K_{eff} and the Neutron Life Cycle; Reactivity and Criticality	Reactor Theory Chapter 1: §1.4 + PDF Files	HW#4 (due 9/27)	**09/15:** Lab #3 - Reactor Startup
5	**09/20 - Quiz #1**: *covering Lectures 1-3, Lab 1, and HW's 1-3.*	***NONE***	***NONE***	**09/22: No Lab**
6	**09/27 – Lect. #5:** Cyber Informed Engineering for Nuclear Systems	PDF Files	HW#5 (due 10/4)	**09/29:** Lab #3 - Reactor Startup
7	**10/04 – Lect. #6:** Sub-Critical Multiplication & 1/M Plots	Reactor Theory Chapter 1: §1.5 + PDF Files	HW#6 (due 10/13)	**10/06:** Lab #4 - 1/M Approach to Criticality
8	***10/11 – Fall Break – No Class – Lecture moved to Thursday 10/13***	***NONE***	***NONE***	**10/13 – No Lab**
9	**10/18 – Lect. #7:** Delayed –vs- Prompt Neutrons and β_{eff}	Reactor Theory Chapter 2: §2.1-§2.3, + PDF Files	HW#7 (due 10/25)	**10/20:** Lab #5 - Control Rod Calibration
10	**10/25 – Quiz #2**: *covering Lectures 4-6, Labs 3-4, and HW's 4-6*	Reactor Theory Chapter 2: §2.4–§2.6 + PDF Files	HW#8 (due 11/1)	**10/27: Lect. #8**: Reactor Period, InHour Equation; Exponential Power Equations
11	**11/01 – Lect. #9:** Reactivity Coefficients (Moderator Temperature, Voids, Power Defect & Fission Product Poisons)	Reactor Theory Chapter 2: §2.7-§2.9.3 + PDF Files	HW#9 (due 11/8)	**11/03:** Lab #6 – Power Defect Lab
12	**11/08 – Lect. #10**: Reactivity Balances, Shutdown Margin	Reactor Theory Chapter 2: § 2.8-§2.13 + PDF Files	HW#10 (due 11/15)	**11/10:** Lab #7 - Axial Power Profile
13	**11/15 – Lect. #11**: Neutron Flux and Power Density; Heat Balance & Power Calibrations	Reactor Theory Chapter 3: §3.1-§3.7 + PDF Files	HW#11 (due 11/22)	**11/17:** Lab #8 - Heat Balance & Power Calibration
14	**11/22 – Special Topic:** Chernobyl Accident			**11/24: Thanksgiving Break**
15	**11/29 (*last day of classes*) – Review**			
	Final Exam: 12/13/2022, 8:30-11:00AM			

*All dates for labs are tentative based on reactor availability